

Reads over Atom Spaces: A Mathematical View of Context, Retrieval, and Parameter Routing in Language Models

Farshid Mossaiby^{a,*}

^aDepartment of Civil Engineering, University of Isfahan, 81744-73441, Isfahan, Iran

Abstract

Finite attention, external retrieval, and parameter-expert routing all use query-conditioned weighted reads, despite differing substantially in their internal computation and training objectives. We provide a common mathematical vocabulary for these mechanisms through valued atom spaces $(\Omega, \mathcal{F}, \nu, \nu)$ and a Bochner read operator $R(x; \Omega) = \int_{\Omega} v(\omega) d\mu_x(\omega)$. In finite neural implementations, this read specializes to a softmax-weighted combination of value vectors. Under this framework, we analyze a Lipschitz-margin condition sufficient for newly inserted atoms to enter an exact top- k set under an explicit global premise, prove exact ranked top- k truncation bounds exhibiting exponential tail decay with an error-budgeted stopping rule, and formalize signed reference measures where anti-atoms algebraically cancel external-read contributions alongside perturbation bounds. The accompanying studies are mechanism diagnostics rather than a claim of a new general-purpose language-model architecture. A finite-dataset Lipschitz diagnostic is vacuous despite perfect synthetic top-1 entry, and exact tail certification requires all scores. On CounterFact-500, held-out ROC calibration ($\tau^* = 0.55$, calibration AUC = 0.817) raises activated paraphrase recall from 38.2% to 53.2%, while neighborhood false activation rises from 9.6% to 16.1%. Five-seed synthetic routing exposes corpus-to-parameter confusion, and matched signed pairs cancel only the external-read contribution, with cancellation degrading rapidly under key mismatch. A direct target-embedding intervention in GPT-2 improves first-token CounterFact scores on $N = 50$ records but does not constitute an end-to-end comparison with retrieval-augmented generation or model editors. The framework

*Corresponding author.

Email address: mossaiby@eng.ui.ac.ir (Farshid Mossaiby)

clarifies shared structure and its operational limits, and is positioned relative to RAG and memory-based editing methods.

Keywords: Language models, Atom spaces, Measure theory, External memory, Vector retrieval, Expert routing, Model editing

1. Introduction

Autoregressive language models encode linguistic abstractions, syntax, and factual knowledge within dense parameter matrices [1, 2, 3]. While gradient-based optimization successfully acquires structural linguistic generalizations, compiling dynamic factual statements directly into static weights presents persistent operational challenges. Real-world facts obsolesce rapidly, require targeted localized updates, and are subject to regulatory erasure mandates (e.g., the “Right to be Forgotten”). Consequently, post-hoc weight modification often triggers catastrophic interference, parameter drift, or high computational costs [4, 5].

To address the rigidity of dense parameter storage, multiple architectural strategies have emerged across distinct sub-disciplines. In-context reasoning caches dynamic token representations directly within the key-value sequence memory of the model [1]. In parallel, retrieval-augmented generation (RAG) and nearest-neighbor language models (k NN-LM) decouple memory from parameters by querying external vector databases of textual passages [6, 7, 8]. Meanwhile, modular computation mechanisms, including Mixture-of-Experts (MoE) architectures and parameter-efficient adapter banks, dynamically route tokens across banks of specialized sub-networks [9, 10, 11]. Although these three paradigms all rely fundamentally on key-value similarity matching [3, 12, 13], they are conventionally formulated using disconnected notations, disparate routing heuristics, and distinct operational semantics.

In this work, we present a unified mathematical formulation for in-context attention, external retrieval, and parameter-expert output routing. By expressing each mechanism through an integral of a query-conditioned probability measure against a value map over a valued atom space, we establish a shared analytical umbrella. In discrete neural architectures, this Bochner read operator specializes algebraically to query-conditioned softmax-weighted convex combinations over value representations.

Through this lens, we establish a unified analytical foundation and examine its operational properties. First, we define the read operator $R(x; \Omega)$ over valued atom spaces $(\Omega, \mathcal{F}, \nu, \nu)$ and formulate an integrated tri-space architecture that co-

32 ordinates among context (Ω_{ctx}), corpus retrieval (Ω_{corpus}), and parameter experts
 33 (Ω_{params}). Building upon this operator, we derive a checkable Lipschitz-margin
 34 condition sufficient for a novel atom to enter an exact top- k retrieved set under
 35 an explicit global metric premise, pairing this result with finite-dataset diagnos-
 36 tics that demonstrate its conservatism in high-dimensional semantic corpora. To
 37 address computational tractability in large-scale retrieval, we prove Theorem 1,
 38 establishing two exponential tail bounds on exact ranked top- k reads, and formu-
 39 late Corollary 1 as an error-budgeted stopping rule that certifies numerical trun-
 40 cation error. Extending the reference measure to signed spaces ($v = v^+ - v^-$),
 41 we formalize the exact algebraic cancellation of external-read contributions via
 42 matched anti-atoms, accompanied by an analytical perturbation bound character-
 43 izing stability under key and value mismatches. Finally, across five random seeds,
 44 we evaluate the framework through controlled mechanism studies encompassing
 45 templated cumulative insertion, surface-matched tri-space routing, approximate
 46 cancellation stress tests, CounterFact retrieval with HNSW candidate indexing and
 47 ROC threshold calibration, and exploratory GPT-2 factual steering baselines.

48 2. From a Discrete Function to an Integral

49 2.1. The Token-Level Function

50 A language model is conventionally defined as a conditional probability dis-
 51 tribution over a finite vocabulary V of size d , given an input sequence. Fixing a
 52 maximum context capacity c and introducing a null padding symbol $\emptyset$ yields an
 53 augmented vocabulary $\bar{V} = V \cup \{\emptyset\}$. An input sequence is represented as a tuple

$$x = (x_1, \dots, x_c) \in \bar{V}^c,$$

54 where unassigned sequence positions are padded with $\emptyset$. A language model is a
 55 mapping

$$p : \bar{V}^c \rightarrow \Delta^{d-1}, \quad \Delta^{d-1} = \left\{ z \in \mathbb{R}^d : z_i \geq 0, \sum_{i=1}^d z_i = 1 \right\}.$$

56 2.2. Attention as an Integral

57 Let $\phi : \bar{V} \rightarrow \mathbb{R}^e$ denote an embedding map with $\phi(\emptyset) = 0$. Extending the
 58 sequence to a step function over the continuous interval $[0, c]$ gives:

$$f_x(t) = \phi(x_{\lceil t \rceil}), \quad t \in (0, c],$$

59 with $f_x(0) = \phi(x_1)$. Let $\alpha_x : [0, c] \rightarrow \mathbb{R}_{\geq 0}$ be an attention weight density satisfy-
 60 ing $\int_0^c \alpha_x(t) dt = 1$, with $\alpha_x(t) = 0$ whenever $x_{[t]} = \emptyset$. The context representation
 61 vector is:

$$z(x) = \int_0^c \alpha_x(t) f_x(t) dt \in \mathbb{R}^e,$$

62 and the corresponding token prediction distribution is $p(x) = \text{softmax}(Wz(x) +$
 63 $b)$. Because f_x is piecewise constant over unit intervals, the continuous integral
 64 evaluates algebraically to the standard discrete attention weighted sum [1, 14]:

$$\int_0^c \alpha_x(t) f_x(t) dt = \sum_{i=1}^c \left(\int_{i-1}^i \alpha_x(t) dt \right) \phi(x_i) = \sum_{i=1}^c \alpha_i \phi(x_i).$$

65 This continuous formulation naturally generalizes by replacing the sequence index
 66 interval $[0, c]$ with an arbitrary measurable *atom space*.

67 3. Atom Spaces: A Common Object for Context, Retrieval, and Parameters

68 **Definition 1** (Valued Atom Space). A valued atom space is a tuple $(\Omega, \mathcal{F}, \nu, v)$,
 69 where $(\Omega, \mathcal{F})$ is a measurable space of elements designated as atoms, ν is a σ -finite
 70 reference measure, and $v : \Omega \rightarrow \mathbb{R}^e$ is a measurable value map into a common
 71 output representation space.

72 **Definition 2** (Query-Conditioned Kernel and Read Operator). Given a query $x \in$
 73 $\mathcal{X}$, let $K_x : \Omega \rightarrow \mathbb{R}_{\geq 0}$ be measurable with

$$0 < Z_x := \int_{\Omega} K_x(\omega) d\nu(\omega) < \infty.$$

74 Then K_x induces a query-conditioned probability measure μ_x on $(\Omega, \mathcal{F})$ via the
 75 Radon–Nikodym derivative:

$$d\mu_x(\omega) = \frac{K_x(\omega)}{Z_x} d\nu(\omega).$$

76 If $\int_{\Omega} \|v(\omega)\|_2 d\mu_x(\omega) < \infty$, the read conditioned on x is the Bochner integral:

$$R(x; \Omega) = \int_{\Omega} v(\omega) d\mu_x(\omega) \in \mathbb{R}^e.$$

Table 1: Three structural realizations of the unified read operator $R(x; \Omega)$.

Instance	Atom Space Ω	Atom ω	Value Map $v(\omega)$	Operational Equivalent
Context	$\{1, \dots, c\}$	Token index	Embedding $\phi(x_\omega)$	In-context self-attention [1]
Corpus	$\mathcal{D}$ (mutable)	Factual record	Encoded fact vector	Dense retrieval / kNN-LM [7]
Parameters	Θ (expert bank)	Modular expert	Expert output $g_\omega(x) \in \mathbb{R}^e$	MoE-style output routing [9, 10]

When ν is a discrete counting measure over a finite set $\Omega = \{\omega_1, \dots, \omega_N\}$, the Bochner integral evaluates directly to a softmax-weighted sum:

$$R(x; \Omega) = \sum_{i=1}^N \frac{K_x(\omega_i)}{\sum_{j=1}^N K_x(\omega_j)} v(\omega_i).$$

As summarized in Table 1, in-context attention, dense retrieval, and expert-output routing are structural realizations of this read operator, differing in their domain of integration and the query-dependence of their values.

3.1. The Unified Architecture

Context, external corpus memory, and parameter routing combine into an integrated representation:

$$z(x) = \lambda_{\text{ctx}}(x)R(x; \Omega_{\text{ctx}}) + \lambda_{\text{corpus}}(x)R(x; \Omega_{\text{corpus}}) + \lambda_{\text{params}}(x)R(x; \Omega_{\text{params}}),$$

$$p(x) = \text{softmax}(W_0 z(x) + b_0),$$

where $\lambda(x) \in \Delta^2$ is an input-dependent gating distribution across the three spaces, and $W_0 \in \mathbb{R}^{d \times e}$, $b_0 \in \mathbb{R}^d$ represent the linguistic projection head. For parameter experts, the value is query-dependent, $v_x(\omega) = g_\omega(x)$; the read is therefore an output-space mixture, not an average of parameter tensors.

3.2. System, Calibration, and Tying Principles

Translating the tri-space read formulation into a functional architecture requires addressing scale calibration and representation tying. Because the cardinalities of the respective spaces vary across orders of magnitude—ranging from $|\Omega_{\text{ctx}}| \sim 10^3$ tokens in context and $|\Omega_{\text{params}}| \sim 10^1$ modular experts to $|\Omega_{\text{corpus}}| \sim 10^9$ external passages—normalizing a single joint softmax across their disjoint union $\Omega_{\text{ctx}} \sqcup \Omega_{\text{corpus}} \sqcup \Omega_{\text{params}}$ would cause the most populous space to artificially dominate probability mass. To preserve balanced access, separate intra-space measure normalization is coupled with calibrated convex gating $\lambda(x)$. Crucially,

key representations across distinct spaces must share identical geometric scales through unit L_2 -normalization, which prevents differing dot-product magnitudes from biasing the router toward unnormalized representations.

Furthermore, for token generation tasks, the external read vector $z_{\text{corpus}}(x)$ must project coherently onto candidate vocabulary tokens. Letting $E \in \mathbb{R}^{d \times e}$ denote the vocabulary embedding matrix, we tie atom values via $v(\omega) = E[y_\omega]/\|E[y_\omega]\|$ and evaluate final logits as Ez/τ . Under this parameter tying, isolating an atom ω^* aligns $z(x)$ directly with the corresponding target row of the vocabulary matrix.

3.3. Signed Atom Spaces and the Hahn–Jordan Decomposition

To support zero-retraining cancellation in external memory, we extend the reference measure ν to a *signed measure*. By the Hahn–Jordan Decomposition Theorem, any signed reference measure uniquely decomposes into mutually singular non-negative measures:

$$\nu = \nu^+ - \nu^-, \quad \nu^+ \perp \nu^-,$$

where ν^+ governs positive constructive atoms and ν^- governs negative “anti-atoms”. Defining the total variation measure $|\nu| = \nu^+ + \nu^-$, we assume $0 < Z_x^\pm = \int_\Omega K_x d|\nu| < \infty$. The normalized signed kernel-weighted measure is:

$$d\mu_x^\pm(\omega) = \frac{K_x(\omega)}{Z_x^\pm} (d\nu^+(\omega) - d\nu^-(\omega)).$$

Its total variation is normalized to unity, while its signed total mass varies in $[-1, 1]$; unlike μ_x , it is not a probability measure.

4. Mathematical Conditions for Post-Hoc Knowledge Editing

Let the query-key kernel be a scaled inner product:

$$K_x(\omega) = \exp\left(\frac{\langle \phi_q(x), \phi_k(\omega) \rangle}{\tau}\right),$$

where $\phi_q : \mathcal{X} \rightarrow \mathbb{R}^m$ and $\phi_k : \Omega \rightarrow \mathbb{R}^m$ are query and key encoders, and $\tau > 0$ is a temperature parameter. Base training optimizes ϕ_q and ϕ_k on an initial pool $\Omega_{\text{train}} \subset \Omega$. Post-hoc editing requires that inserting a novel atom $\omega^* \notin \Omega_{\text{train}}$ at test time reliably captures probability mass $\mu_x(\omega^*)$ for associated queries x^* , without modifying encoder parameters.

124 4.1. Kernel Locality

125 **Definition 3** (Metric Locality). *The key encoder ϕ_k is L -Lipschitz continuous with*
 126 *respect to a semantic pseudo-metric d_Ω on Ω if*

$$\|\phi_k(\omega) - \phi_k(\omega')\|_2 \leq L d_\Omega(\omega, \omega') \quad \forall \omega, \omega' \in \Omega.$$

127 **Proposition 1** (Score Stability on Unseen Atoms). *Let ϕ_k be L -Lipschitz with*
 128 *respect to d_Ω . For any training anchor $\omega_0 \in \Omega_{\text{train}}$ and any newly inserted atom*
 129 *$\omega^* \in \Omega$:*

$$\left| \langle \phi_q(x), \phi_k(\omega^*) \rangle - \langle \phi_q(x), \phi_k(\omega_0) \rangle \right| \leq \|\phi_q(x)\|_2 L d_\Omega(\omega^*, \omega_0).$$

130 *Proof.* By the Cauchy–Schwarz inequality and the Lipschitz condition:

$$\begin{aligned} \left| \langle \phi_q(x), \phi_k(\omega^*) - \phi_k(\omega_0) \rangle \right| &\leq \|\phi_q(x)\|_2 \|\phi_k(\omega^*) - \phi_k(\omega_0)\|_2 \\ &\leq \|\phi_q(x)\|_2 L d_\Omega(\omega^*, \omega_0). \end{aligned} \quad \square$$

131 4.2. The Retrieval Margin Condition

132 Let $\tau_k(x) = \max_{\omega \in \Omega_{\text{train}}}^{(k)} \langle \phi_q(x), \phi_k(\omega) \rangle$ denote the k -th highest score among ex-
 133 isting training atoms. For the novel atom ω^* to enter the top- k retrieved set under
 134 query x , it must satisfy $\langle \phi_q(x), \phi_k(\omega^*) \rangle > \tau_k(x)$. Combining this with Proposi-
 135 tion 1 yields a checkable sufficient condition:

$$\langle \phi_q(x), \phi_k(\omega_0) \rangle - \|\phi_q(x)\|_2 L d_\Omega(\omega^*, \omega_0) > \tau_k(x), \quad (1)$$

136 where ω_0 is a training anchor semantically proximal to ω^* . Equation (1) is a suffi-
 137 cient condition for top- k inclusion; successful prediction additionally depends on
 138 retrieved probability mass, value representation, output margin, and specificity on
 139 controls.

140 *Finite-Dataset Diagnostic and Conservatism of the Margin Condition.* We diag-
 141 nose Equation (1) on the finite benchmark. We instantiate $d_\Omega(\omega, \omega') = \|\phi_0(\omega) -$
 142 $\phi_0(\omega')\|_2$ using the Euclidean distance between frozen MiniLM backbone em-
 143 beddings. For the residual adapter $\phi(x) = \text{Normalize}(x + 0.1 \cdot \text{MLP}(x))$, an
 144 operator-norm calculation combined with the minimum adapted norm observed
 145 on this dataset gives a finite-dataset estimate $L \leq 2.105$, while the maximum ob-
 146 served pairwise directional ratio is $\hat{L} = 2.236$. Neither quantity establishes the
 147 global premise of Proposition 1; they are diagnostics of its conservatism on the
 148 sampled records. Evaluating Equation (1) across all 40 insertion facts using their

149 nearest training anchors ω_0 reveals a stark contrast between observed entry and
 150 this diagnostic: actual top-1 entry is 100.0%, whereas the margin is positive for
 151 0.0% of records across top-1, top-5, and top-10. Cauchy–Schwarz permits ad-
 152 versarial alignment in 384 dimensions; with inter-fact semantic distances averag-
 153 ing $d_\Omega \approx 0.78$, the penalty $\hat{L} d_\Omega \approx 1.74$ overwhelms the observed score margin
 154 (LHS ≈ -1.34 versus $\tau_k \approx 0.35$). Thus, the experiment illustrates that a global
 155 worst-case condition can be vacuous on sparse semantic corpora; non-vacuous
 156 guarantees would require independently established local or directional bounds.

157 4.3. Anti-Atoms and Exact External-Read Cancellation

158 **Proposition 2** (Matched-Pair Cancellation). *Let an atom $\omega^* \in \Omega_{\text{corpus}}$ encode*
 159 *a target fact with tied token value $v^* = E[y^*]/\|E[y^*]\|$. If an anti-atom ω^- is*
 160 *inserted into the signed atom space with identical key $\phi_k(\omega^-) = \phi_k(\omega^*)$, identical*
 161 *value $v^- = v^*$, and equal positive masses $m^* := v^+(\{\omega^*\}) = m^- := v^-(\{\omega^-\}) >$*
 162 *0, then for any query x :*

$$R^\pm(x; \{\omega^*, \omega^-\}) = \mathbf{0}.$$

163 *Proof.* Under the signed measure integral of Section 3.3:

$$\int_{\{\omega^*, \omega^-\}} v(\omega) d\mu_x^\pm(\omega) = \frac{K_x(\omega^*)m^*v^* - K_x(\omega^-)m^-v^*}{Z_x^\pm} = \frac{K_x(\omega^*)m^*(v^* - v^*)}{Z_x^\pm} = \mathbf{0}.$$

164 This establishes exact cancellation of the matched pair’s contribution to the signed
 165 external read. $\square$

166 Proposition 2 does not imply that the complete model representation is zero:
 167 background atoms, context, parameter experts, residual model states, and output
 168 bias remain. The positive record and anti-atom remain stored, and the proposition
 169 does not claim removal of knowledge compiled into dense pretrained parameters.

170 **Proposition 3** (Approximate Pair Cancellation). *Let $a = K_x(\omega^*)v^+(\{\omega^*\})$, $b =$*
 171 *$K_x(\omega^-)v^-(\{\omega^-\})$, and let $Z_x^\pm > 0$ be the total-variation normalization. For posi-*
 172 *tive value v and anti-atom value $\tilde{v}$, the residual pair contribution satisfies:*

$$\left\| \frac{av - b\tilde{v}}{Z_x^\pm} \right\|_2 \leq \frac{|a - b| \|v\|_2 + b \|v - \tilde{v}\|_2}{Z_x^\pm}.$$

173 *Proof.* Write $av - b\tilde{v} = (a - b)v + b(v - \tilde{v})$ and apply the triangle inequality, then
 174 divide by $Z_x^\pm$. $\square$

175 Exact cancellation is sensitive to key-induced kernel mismatch, unequal mass,
 176 and value mismatch. Paraphrases affect cancellation only through their effect on
 177 kernel weights; identical keys cancel for all queries.

178 5. Computational Tractability and the Softmax Tail Truncation Bound

179 Evaluating $R(x; \Omega) = \int_{\Omega} v(\omega) d\mu_x(\omega)$ densely over a large corpus is compu-
 180 tationally expensive. We analyze numerical truncation to the exact ranked top- k
 181 elements:

$$\hat{R}_k(x; \Omega) = \sum_{i=1}^k v(\omega_{(i)}) \frac{K_x(\omega_{(i)})}{\sum_{j=1}^k K_x(\omega_{(j)})}.$$

182 In this discrete setting, numerical truncation corresponds to numerical quadra-
 183 ture of the query-conditioned measure against the value map, omitting the low-
 184 probability tail $Z_{\text{tail}} = \sum_{i=k+1}^N e^{s_{(i)}/\tau}$. Dense scoring over N items costs $\mathcal{O}(Ne)$
 185 before the top- k read. When paired with an ANN index, aggregation costs:

$$\mathcal{O}(c \cdot e) + \mathcal{O}(k_{\text{corpus}} \cdot e) + \mathcal{O}(k_{\text{params}} \cdot e).$$

186 Let $N = |\Omega|$, and $V_{\max} := \sup_{\omega \in \Omega} \|v(\omega)\|_2 < \infty$. For query x , let $s_i :=$
 187 $\langle \phi_q(x), \phi_k(\omega_i) \rangle$, with ordering $s_{(1)} \geq \dots \geq s_{(N)}$. Define $Z := \sum_{j=1}^N e^{s_{(j)}/\tau}$ and
 188 $Z_k := \sum_{j=1}^k e^{s_{(j)}/\tau}$. Define the retrieval spectral gap at rank k as $\Delta_k(x) := s_{(1)} -$
 189 $s_{(k+1)} \geq 0$.

190 **Theorem 1** (Softmax Tail Truncation Error Bound). *For any query x and trun-*
 191 *cation depth $k \in \{1, \dots, N-1\}$, the Euclidean approximation error of top- k*
 192 *truncation satisfies:*

$$\|R(x; \Omega) - \hat{R}_k(x; \Omega)\|_2 \leq 2V_{\max} \cdot (N - k) \cdot \exp\left(-\frac{\Delta_k(x)}{\tau}\right).$$

193 *An alternative bound is:*

$$\|R(x; \Omega) - \hat{R}_k(x; \Omega)\|_2 \leq 2V_{\max} \cdot \left(\frac{N - k}{k}\right) \cdot \exp\left(-\frac{s_{(k)} - s_{(k+1)}}{\tau}\right).$$

194 *Proof.* Decompose the difference between the full read and the top- k truncation:

$$\begin{aligned} R(x; \Omega) - \hat{R}_k(x; \Omega) &= \sum_{i=1}^k v_{(i)} e^{s_{(i)}/\tau} \left(\frac{1}{Z} - \frac{1}{Z_k} \right) + \sum_{i=k+1}^N v_{(i)} \frac{e^{s_{(i)}/\tau}}{Z} \\ &= \frac{Z_k - Z}{Z Z_k} \sum_{i=1}^k v_{(i)} e^{s_{(i)}/\tau} + \frac{1}{Z} \sum_{i=k+1}^N v_{(i)} e^{s_{(i)}/\tau}. \end{aligned}$$

195 Let $Z_{\text{tail}} := Z - Z_k = \sum_{i=k+1}^N e^{s(i)/\tau}$. Notice $\frac{1}{Z_k} \sum_{i=1}^k v_{(i)} e^{s(i)/\tau} = \hat{R}_k(x; \Omega)$, and
 196 define $R_{\text{tail}} := \frac{1}{Z_{\text{tail}}} \sum_{i=k+1}^N v_{(i)} e^{s(i)/\tau}$. Substituting yields:

$$R(x; \Omega) - \hat{R}_k(x; \Omega) = \frac{Z_{\text{tail}}}{Z} (R_{\text{tail}} - \hat{R}_k(x; \Omega)).$$

197 Taking norms and applying the triangle inequality:

$$\|R(x; \Omega) - \hat{R}_k(x; \Omega)\|_2 \leq \frac{Z_{\text{tail}}}{Z} (\|R_{\text{tail}}\|_2 + \|\hat{R}_k(x; \Omega)\|_2) \leq 2V_{\max} \frac{Z_{\text{tail}}}{Z}.$$

198 Since $Z \geq e^{s(1)/\tau}$ and $Z_{\text{tail}} \leq (N - k)e^{s(k+1)/\tau}$:

$$\frac{Z_{\text{tail}}}{Z} \leq \frac{(N - k)e^{s(k+1)/\tau}}{e^{s(1)/\tau}} = (N - k) \exp\left(-\frac{s(1) - s(k+1)}{\tau}\right).$$

199 Substituting proves the first bound. Alternatively, $Z \geq ke^{s(k)/\tau}$ while $Z_{\text{tail}} \leq (N -$
 200 $k)e^{s(k+1)/\tau}$, giving

$$\frac{Z_{\text{tail}}}{Z} \leq \frac{N - k}{k} \exp\left(-\frac{s(k) - s(k+1)}{\tau}\right),$$

201 which proves the second bound. The second bound is tighter whenever $s(1) - s(k) <$
 202 $\tau \log k$. $\square$

203 **Corollary 1** (Error-Budgeted Exact Truncation). *Let $\rho_k(x) = Z_{\text{tail}}/Z$ be the exact*
 204 *softmax mass outside the first k ranked atoms. For any requested $\varepsilon > 0$, define*

$$k_\varepsilon(x) = \min\{k \in \{1, \dots, N\} : 2V_{\max}\rho_k(x) \leq \varepsilon\}.$$

205 *Then $\|R(x; \Omega) - \hat{R}_{k_\varepsilon(x)}(x; \Omega)\|_2 \leq \varepsilon$.*

206 *Proof.* By the proof of Theorem 1, $\|R - \hat{R}_k\|_2 \leq 2V_{\max}\rho_k(x)$. Substituting the
 207 condition for $k_\varepsilon(x)$ yields the result. $\square$

208 Theorem 1 bounds truncation error once score ordering is known. Comput-
 209 ing the certificate requires all scores, making it a certification rule rather than a
 210 sublinear algorithm.

211 6. Empirical Evaluation

212 We examine inserted-record retrieval, adaptive exact top- k truncation, Coun-
 213 terFact retrieval with HNSW candidates and ROC threshold calibration, cumula-
 214 tive insertion, tri-space routing, signed-read suppression under perturbation, and
 215 GPT-2 steering.

216 6.1. Experimental Setup

217 *Corpus and Domain Design.* We construct a benchmark of 200 structured techni-
 218 cal facts distributed across four domains: *Astronomy & Astrophysics*, *Systems &*
 219 *Computing*, *Chemistry & Materials Science*, and *Molecular Biology & Genetics*.
 220 The first 40 items per domain form the adapter-training pool; the remaining 10
 221 items per domain serve as post-hoc insertions. Their labels remain in the global
 222 output vocabulary, making this a transductive label-space evaluation.

223 *Encoder Architecture and Inductive Bias.* We employ a frozen sentence trans-
 224 former (sentence-transformers/all-MiniLM-L6-v2, $d_{\text{in}} = 384$) as the ge-
 225 ometric backbone, paired with a metric-preserving residual adapter:

$$\phi(x) = \text{Normalize}(x + 0.1 \cdot \text{MLP}(x)),$$

226 where the final MLP layer is zero-initialized. Values are tied to token embeddings:
 227 $v(\omega) = E[y_\omega] / \|E[y_\omega]\|$, with predictions evaluated as $p(y | x) \propto \langle z(x), E[y] \rangle / \tau$.

228 *Reproducibility Environment.* Artifacts were generated using Python 3.11.9, Py-
 229 Torch 2.6.0+cu124, NumPy 2.4.6, sentence-transformers 5.7.0, and transformers
 230 4.48.3 on an NVIDIA GeForce RTX 3050 Ti Laptop GPU (4 GB VRAM) with
 231 an AMD Ryzen 7 5800U host. Pinned versions are listed in `requirements.txt`.
 232 The routing sweep reports seeds {40, 41, 42, 43, 44}.

233 6.2. Post-Hoc Knowledge Editing Results

234 Table 2 compares the trained read with frozen MiniLM exact 1-nearest-
 235 neighbor retrieval across pool densities $\Omega_{\text{train}} \in \{1, 3, 10, 40\}$ items per domain.

236 Across 40 insertion records and all density conditions, both methods obtain
 237 100% target classification, with no argmax changes on active training queries. The
 238 frozen exact-retrieval baseline matches the trained read: this templated benchmark
 239 does not demonstrate that the residual adapter improves retrieval over raw embed-
 240 dings. At density 40, the maximum output probability shift is 2.86×10^{-6} .

241 As training density scales from 1 to 40 atoms per domain, the product of linear-
 242 layer spectral norms increases from 0.469 to 1.003, while target attention mass
 243 decreases from 99.97% to 99.75%.

244 6.3. Convergence of Exact Ranked Top-k Truncation

245 Table 3 presents empirical truncation error $\|R_{\text{exact}}(x) - \hat{R}_k(x)\|_2$ across temper-
 246 atures and cutoffs k on the largest atom pool (density 40, 160 active background
 247 atoms).

Table 2: Inserted-record retrieval across 200 templated facts. Efficacy and generality denote target classification for canonical queries and paraphrases. Specificity denotes unchanged argmax predictions on active training queries.

Metric	1 / topic	3 / topic	10 / topic	40 / topic
<i>Trained Residual Read</i>				
Efficacy (Target Recall on ω^*)	100.0%	100.0%	100.0%	100.0%
Generality (Paraphrase Query)	100.0%	100.0%	100.0%	100.0%
Specificity (Neighborhood Retention)	100.0%	100.0%	100.0%	100.0%
<i>Frozen MiniLM Exact 1-Nearest-Neighbor Baseline</i>				
Efficacy (Target Recall on ω^*)	100.0%	100.0%	100.0%	100.0%
Generality (Paraphrase Query)	100.0%	100.0%	100.0%	100.0%
Specificity (Argmax Retention)	100.0%	100.0%	100.0%	100.0%
<i>Representation and Metric Dynamics</i>				
Novel Atom Attention Mass $\mu_x(\omega^*)$	99.97%	99.97%	99.94%	99.75%
Adapter Linear Spectral Product	0.469	0.818	0.859	1.003

At density 40, $\tau = 0.10$, and $k = 1$, the maximum observed error is 0.0244, and the smaller pointwise bound covers 100% of queries with a minimum slack of 0.0923. Truncation dynamics exhibit two sharply distinct regimes depending on temperature. At low temperatures ($\tau = 0.05$), probability mass concentrates heavily on the nearest record, yielding a mean top-1 error of 3.82×10^{-6} that rapidly diminishes to 1.41×10^{-6} at $k = 20$. Conversely, at elevated temperatures ($\tau \geq 0.50$), probability mass diffuses broadly across background atoms, causing truncation errors to approach 1.0 when cut off prematurely.

Adaptive error budgets and scaling. Evaluating Corollary 1 on 64 correlated synthetic queries with random keys and values in $\mathbb{R}^{128}$ for $N \in \{1000, 5000, 10000, 25000\}$ confirms 100% empirical coverage across $\varepsilon \in \{0.01, 0.05, 0.10\}$. However, the certificate is conservative: at $N = 25,000$ and $\varepsilon = 0.05$, mean k_ε is 13,026 (52.1% of the corpus) for $\tau = 0.05$ and 21,467 (85.9%) for $\tau = 0.10$. Exact scoring and sorting take 54.6 and 18.2 ms on the benchmark GPU.

6.4. CounterFact External-Memory Retrieval and Calibrated Activation

We evaluate frozen MiniLM retrieval geometry on the first 500 CounterFact records [4]. Memory keys concatenate the rewrite prompt with its requested target;

Table 3: Softmax tail truncation error $\|R_{\text{exact}} - \hat{R}_k\|_2$ evaluated on context vectors across temperatures τ and cutoffs k . Under unit-normalized embeddings ($V_{\text{max}} = 1$), error is bounded by 2.0 and decays exponentially at sharp operational temperatures ($\tau = 0.05, k = 1 \Rightarrow 3.82 \times 10^{-6}$).

Temperature	Top-1	Top-3	Top-5	Top-10	Top-20
$\tau = 0.05$	3.82×10^{-6}	3.01×10^{-6}	2.62×10^{-6}	2.03×10^{-6}	1.41×10^{-6}
$\tau = 0.10$	0.0182	0.0170	0.0162	0.0145	0.0121
$\tau = 0.50$	0.965	0.700	0.558	0.384	0.245
$\tau = 1.00$	0.988	0.623	0.477	0.327	0.217

Table 4: CounterFact-500 retrieval evaluation comparing uncalibrated ($\tau = 0.60$) and data-calibrated ($\tau^* = 0.55$) thresholds. Threshold τ^* is calibrated via Youden’s J on held-out records 500–999 (AUC = 0.817) and evaluated on records 0–499.

Query Formulation	Exact Top-1 Recall	Activated ($\tau = 0.60$)	Calibrated ($\tau^* = 0.55$)
Rewrite prompts (500)	99.2%	99.2%	99.4%
Paraphrase prompts (1,000)	91.0%	38.2%	53.2%
Neighborhood prompts (1,000)	–	9.6% (false act.)	16.1% (false act.)
ROC AUC (held-out tuning)		0.817	
HNSW Candidate Recall (10 candidates)		99.96%	

265 evaluation covers 500 rewrite prompts, 1000 paraphrases (2 per record), and 1000
 266 neighborhood prompts (2 per record).

267 To address the trade-off where a fixed cosine threshold of 0.60 prematurely
 268 discards valid paraphrases (38.2% retained), we perform a data-driven threshold
 269 calibration. We partition CounterFact into a calibration split of records 500–999
 270 and an evaluation split of records 0–499. On the calibration split, sweeping activa-
 271 tion thresholds across positive queries (rewrites and paraphrases) versus negative
 272 distractors (neighborhood queries) yields an ROC curve with an AUC of 0.817.
 273 Optimizing Youden’s J statistic on the calibration set identifies an optimal oper-
 274 ating threshold of $\tau^* = 0.55$.

275 As shown in Table 4, calibrating the threshold to $\tau^* = 0.55$ raises activated
 276 paraphrase recall from 38.2% to 53.2%, with neighborhood false activation in-
 277 creasing only modestly from 9.6% to 16.1%. FAISS HNSW achieves 99.96%
 278 exact-top-1 candidate recall with 12.7 ms construction and 23.5 ms search over
 279 all 2500 queries.

Table 5: CounterFact-500 ROME-inspired rank-one baseline on GPT-2 (124M) with a key-covariance prior and real subject spans, versus ten-fact naive fine-tuning and ROME-lite mechanism baselines. Metrics and protocols differ across columns; the table is descriptive rather than a direct method ranking.

Metric	Naive FT (10 facts)	ROME-Lite (10 facts)	ROME-Inspired (500 facts)
Efficacy Score	100% (argmax)	100% (argmax)	85.2% (35.6% argmax)
Paraphrase Score	70% (argmax)	40% (argmax)	59.5%
Neighborhood Score / specificity	58.9% (mean)	87.8% (mean)	70.1%
Control-text perplexity ratio	0.982 mean, 1.022 max	0.996 mean, 1.004 max	1.001 mean, 1.291 max
Edit layer(s)	All parameters	Per-record (mostly layer 10)	Single global layer 3

6.5. CounterFact-Scale ROME-Inspired Rank-One Baseline

To provide a scale-matched parametric reference, we implement a covariance-regularized ROME-inspired rank-one baseline [4] on the same 500 CounterFact records: (i) the edit is inserted at the subject span identified via tokenizer offsets; (ii) a single edit layer (layer 3) is selected via a lightweight causal trace averaged over 50 held-out records; and (iii) rank-one updates use a key covariance matrix $C = \mathbb{E}[kk^\top]$ estimated over 1500 held-out texts: $W \leftarrow W + (C^{-1}k)(v^* - k^\top W)^\top / (k^\top C^{-1}k)$. It retains important deviations from the original ROME protocol: one canonical prompt per edit, no multi-template key averaging or KL regularizer, a CounterFact-derived covariance corpus, and first-subtoken rather than full-sequence scores. It is therefore an ablation, not a reproduction or a directly comparable literature baseline.

At CounterFact-500 scale, the covariance-regularized ROME-inspired baseline obtains an 85.2% Efficacy Score (35.6% argmax), 59.5% Paraphrase Score, and 70.1% Neighborhood Score, with control perplexity remaining stable (1.001 mean, 1.291 max). Because its dataset, target scoring, and optimization protocol differ from the ten-fact columns and standard ROME, these figures should be interpreted as a diagnostic reference rather than a direct comparison or a reproduction result.

6.6. Lifelong Cumulative Knowledge Editing Benchmark

We evaluate lifelong memory stability by inserting all 40 held-out facts sequentially and permanently without teardown, expanding the active pool from 161 to 200 atoms.

Across five seeds (Table 6, Figure 1), all earlier insertions remain correctly classified when the 40th record is added, with zero training argmax changes. Target attention mass averages $99.681 \pm 0.014\%$.

Table 6: Cumulative insertion performance across 40 sequential records without teardown. Target argmax recall and training-query argmax agreement remain $100.0 \pm 0.0\%$ across five seeds.

Sequential Edits Inserted	1	5	10	20	30	40
Total Active Pool Size	161	165	170	180	190	200
Cumulative Efficacy (All Past Edits)	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
Cumulative Generality (Paraphrases)	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
Background Specificity (Original 160)	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
Mean Attention Mass $\mu_x(\omega^*)$	99.84%	99.76%	99.46%	99.60%	99.66%	99.70%

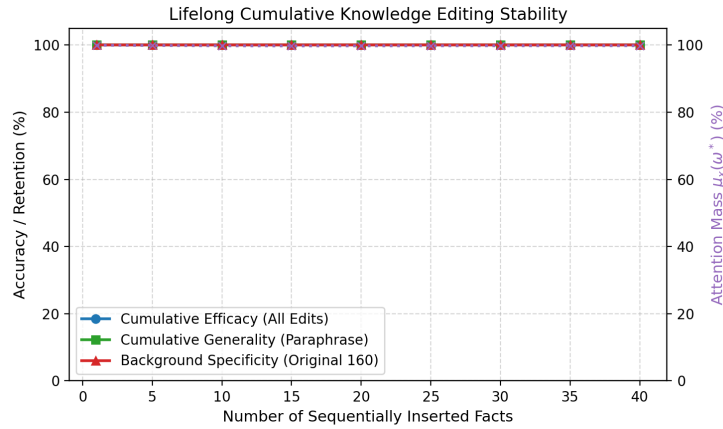


Figure 1: Lifelong cumulative editing stability across 40 sequential fact insertions. Cumulative efficacy, generality, and background specificity remain at 100%, while attention mass $\mu_x(\omega^*)$ stays $> 99.6\%$.

306 6.7. Dynamic Tri-Space Routing and Path Ablation

307 We evaluate the tri-space architecture across three information paths: in-
 308 context reasoning (Ω_{ctx}), corpus world knowledge (Ω_{corpus}), and parameter-resident
 309 expert outputs (Ω_{params}). Router $\lambda(x) \in \Delta^2$ is evaluated on unseen strings using
 310 the shared wrapper “Determine the output for:” across all three families.

311 Table 7 and Figure 2 show route accuracies of 100.0%, 82.4%, and 100.0%, and
 312 task accuracies of 100.0%, 96.8%, and 100.0%. Ablating the intended space re-
 313 duces task accuracy to 0% across all five seeds. Removing explicit route keywords
 314 exposes partial corpus confusion with the parameter path (30.5% gate allocation).

315 6.8. Knowledge Contradiction and Anti-Atom External-Memory Suppression

316 *External-Memory Suppression via Anti-Atoms.* Inserting an anti-atom ω^- with
 317 negative mass -1.0 at the metric coordinate of a sensitive record achieves zero-

Table 7: Five-seed surface-matched routing and ablation (mean $\pm$ sample standard deviation). Route accuracy is the fraction whose largest gate selects the intended space.

Task Domain	Task Acc.	Route Acc.	Mean Gating Distribution $\lambda(x)$			Ablated Acc.
			λ_{ctx}	λ_{corpus}	λ_{params}	
In-Context Reasoning (Ω_{ctx})	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	68.0$\pm$2.1	6.2 $\pm$ 0.9	25.8 $\pm$ 1.3	0.0 $\pm$ 0.0
Corpus World Knowledge (Ω_{corpus})	96.8 $\pm$ 1.8	82.4 $\pm$ 0.9	12.9 $\pm$ 0.7	56.6$\pm$2.2	30.5 $\pm$ 1.7	0.0 $\pm$ 0.0
Parameter Expert Outputs (Ω_{params})	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	3.1 $\pm$ 0.2	3.1 $\pm$ 0.4	93.7$\pm$0.5	0.0 $\pm$ 0.0

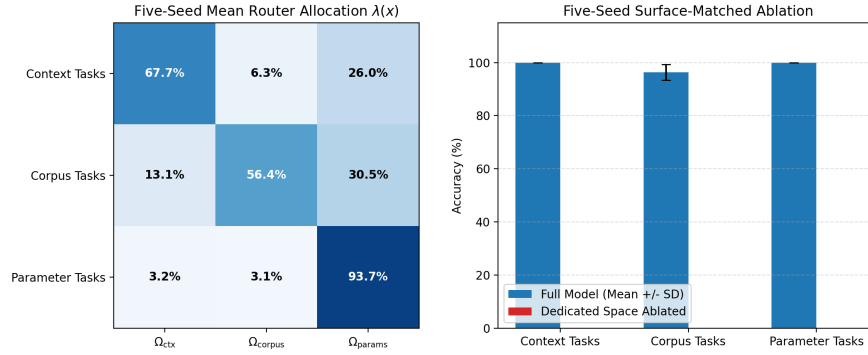


Figure 2: Five-seed surface-matched tri-space evaluation. Left: mean router allocation $\lambda(x)$. Right: mean task accuracy; intended-path ablation reduces accuracy to zero.

318 retraining cancellation in the external read.

319 Table 8 and Figure 3 show target probability dropping from 99.998% to **2.04%**
 320 (1/49), with five-seed mean residual probability of $2.0407 \pm 0.00003\%$. Specificity
 321 on 35 unedited control facts remains **100.0%**.

322 *Approximate cancellation stress test.* Perturbing anti-atom keys, values, masses,
 323 or queries over 50 trials across background sizes 10, 100, and 1000 confirms 100%
 324 analytical bound coverage (3,000 trials). At $N = 100$, key perturbations of 0.01,
 325 0.05, and 0.10 leave mean pair residuals of 4.5%, 72.1%, and 98.0% relative to
 326 the uncanceled contribution. Query perturbations leave numerical-zero residual
 327 because identical keys receive identical kernel weights. Exact anti-atoms are ef-
 328 fective but key mismatch rapidly degrades cancellation.

329 *Competitive Updating.* Inserting contradictory atoms with recency weight $w_{\text{new}} =$
 330 2.5 in an append-only log succeeds on only $44.0 \pm 8.9\%$ of records across five seeds.
 331 In remaining cases, lexical similarity to the original record outweighs the recency
 332 bonus.

Table 8: External-memory suppression and counterfactual updating via signed reads on a 49-token output vocabulary. Matched pairs reduce target probability to $1/|\mathcal{V}| \approx 2.04\%$ in the external read. Specificity on 35 clean background facts remains 100.0%.

Target Entity / Query	Pre-Edit P(Target)	Post-Edit P(Target)	Status	Background Specificity
<i>External-Memory Suppression via Anti-Atoms (unit mass under v^-)</i>				
Secret Base Alpha Coordinates	99.998%	2.04% ($1/ \mathcal{V} $)	Suppressed	100.0%
Patient Confidential Diagnosis	99.998%	2.04% ($1/ \mathcal{V} $)	Suppressed	100.0%
Proprietary Patent Number	99.998%	2.04% ($1/ \mathcal{V} $)	Suppressed	100.0%
Classified Document Location	99.998%	2.04% ($1/ \mathcal{V} $)	Suppressed	100.0%
VIP Cryptographic Key	99.998%	2.04% ($1/ \mathcal{V} $)	Suppressed	100.0%
<i>Counterfactual Competitive Updating ($w_{\text{new}} = 2.5$, Append-Only)</i>				
Ethereum $\rightarrow$ Proof-of-Stake	3.0% (old)	40.0% (new)	Overridden	100.0%
Java $\rightarrow$ Oracle	0.01% (old)	60.1% (new)	Overridden	100.0%
Pluto $\rightarrow$ Dwarf Planet	32.3% (old)	23.9% (new)	Competing	100.0%
Twitter $\rightarrow$ X Corp	99.8% (old)	0.04% (new)	Lexical Bias	100.0%
Python $\rightarrow$ PyPy JIT	98.9% (old)	0.24% (new)	Lexical Bias	100.0%

Table 9: Selected rows from ten-fact greedy GPT-2 evaluation. Factual atoms boost first-token probabilities and correct canonical predictions.

Factual Target Task	Target Probability $P(y^*)$		Ratio	Top Predicted Token	
	Base GPT-2	Atom-Space		Base GPT-2	Atom-Space
QUIC Transport Protocol (UDP)	2.64%	70.00%	26.5 $\times$	the	UDP
Milky Way Black Hole (Sagittarius)	0.47%	64.33%	136.3 $\times$	after	Sag
Safe Systems Language (Rust)	0.03%	32.99%	1116.0 $\times$	a	Rust
Uranium Radiative Decay (lead)	0.87%	9.80%	11.3 $\times$	uranium	lead
CRISPR Bacterial Defense (viruses)	2.25%	38.41%	17.1 $\times$	infect.	viruses

6.9. Autoregressive Generative Modeling with GPT-2

We augment pretrained GPT-2 (124M [15]) by coupling Ω_{corpus} into its final hidden representation with cosine gating (≥ 0.60):

$$\tilde{h}_t = \begin{cases} (1 - \lambda_{\text{read}})h_t + \lambda_{\text{read}}W_{\text{proj}}z_t, & \text{relevant and unconsumed,} \\ h_t, & \text{otherwise.} \end{cases}$$

Table 9 and Figure 4 show first-token argmax correction on 10/10 canonical prompts and 9/10 paraphrases. On ten unrelated control texts, the relevance threshold prevents false activation, keeping perplexity ratio at 1.000. However, multi-token autoregressive decoding reveals degradation once the one-shot token is consumed (e.g., generating “lead, thorium, and lead-based materials”).

Table 10 compares the read operator against naive fine-tuning (FT) and single-layer ROME-lite on the same ten facts. Naive FT achieves 100% canonical efficacy

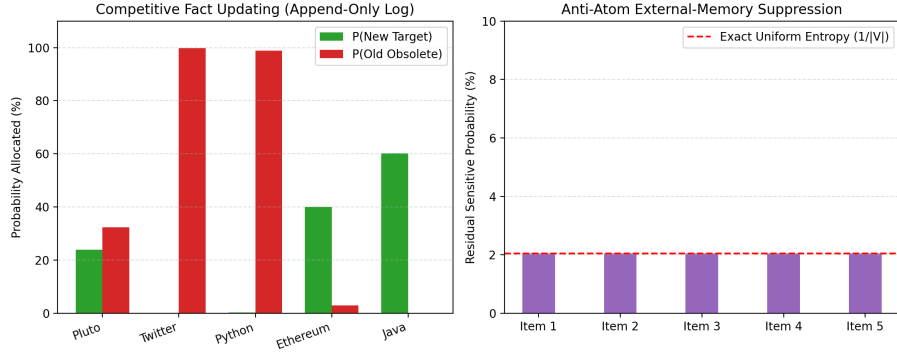


Figure 3: Empirical evaluation of signed atom spaces. Left: Competitive updating in an append-only log ($w_{\text{new}} = 2.5$). Right: anti-atom cancellation drives residual target probability to uniform entropy ($P = 2.04\% \approx 1/|\mathcal{V}|$).

Table 10: Naive full fine-tuning and single-layer rank-one editing (ROME-lite) versus relevance-gated atom-space read on the same ten GPT-2 facts, paraphrases, and controls.

Metric	Atom-Space Read	Naive Fine-Tuning (FT)	ROME-Lite
Canonical efficacy (first-token argmax)	100% (10/10)	100% (10/10)	100% (10/10)
Paraphrase generality (first-token argmax)	90% (9/10)	70% (7/10)	40% (4/10)
Neighborhood specificity (other 9 facts unchanged)	n/a (no weight change)	58.9% (mean)	87.8% (mean)
Control-text perplexity ratio	1.000 (exact)	0.982 mean, 1.022 max	0.996 mean, 1.004 max
Mechanism	Relevance-gated read	Full-parameter AdamW	Rank-one weight update

343 but alters predictions on 41.1% of neighborhood facts. ROME-lite preserves local-
344 ity (87.8% neighborhood retention) but generalizes less effectively to paraphrases
345 (40% vs. 90%).

346 *Expanded CounterFact Generative Steering ($N = 50$ Records).* We scale this
347 direct-intervention mechanism study to $N = 50$ CounterFact records under the
348 calibrated threshold $\tau^* = 0.55$. Across 2,000 bootstrap resamples, base GPT-2
349 assigns $P(\text{new}) > P(\text{true})$ on 16.0% [6.0%, 26.0%] of rewrite prompts and 22.0%
350 [12.0%, 34.0%] of paraphrases, with 0.0% target argmax accuracy. Augmenting
351 GPT-2 with the atom space Ω_{corpus} achieves 100.0% rewrite activation and 58.0%
352 paraphrase activation. Steered GPT-2 achieves **100.0%** [100.0%, 100.0%] rewrite
353 efficacy (100% argmax accuracy) and a paraphrase score of **66.0%** [52.0%, 78.0%]
354 (56.0% exact argmax accuracy, [42.0%, 70.0%]). Since each memory value is the
355 target token embedding and the read is mixed directly into the final hidden state,
356 this demonstrates first-token controllability conditional on retrieval, not an end-to-
357 end RAG or model-editing comparison.

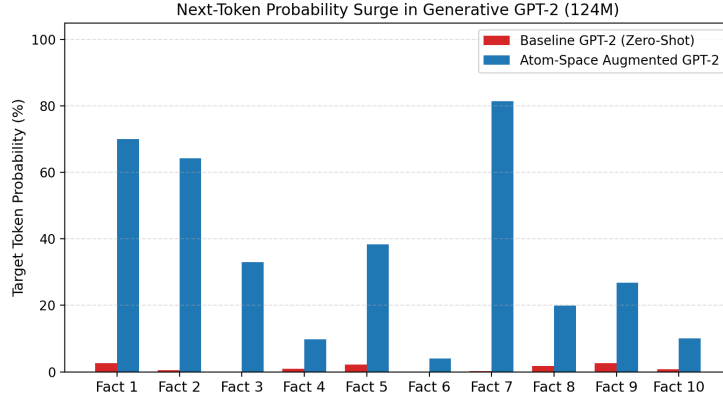


Figure 4: First-token target probabilities in GPT-2 (124M) before and after target-embedding intervention.

7. Discussion and Related Work

The computational elements unified under our framework have extensive histories across distinct areas of machine learning. External corpus reads $R(x; \Omega_{\text{corpus}})$ correspond to nearest-neighbor language models and retrieval-augmented architectures, such as k NN-LM [7, 16], RETRO [8], and memory network designs [17, 12]. Parameter-expert reads $R(x; \Omega_{\text{params}})$ encapsulate modular computation, including sparsely gated Mixture-of-Experts [9, 18, 11] and dynamic adapter routing banks [10]. Finally, standard self-attention [1] constitutes an in-context read $R(x; \Omega_{\text{ctx}})$ over sequence coordinates, whose continuous formulation relates naturally to kernel attention smoothing and linear attention mechanisms [13, 19].

7.1. Relation to Memory-Based and Non-Parametric Model Editing

The atom-space read operator connects directly to non-parametric, semi-parametric, and memory-augmented model editing strategies.

Semi-Parametric Edit Retrieval and Correction (SERAC) [20] caches edits externally and employs an auxiliary classifier to route queries between a base language model and a separate counterfactual sequence-to-sequence model. RoAS shares the objective of keeping base weights frozen, but conceptualizes retrieval as a query-conditioned convex combination $R(x; \Omega)$ that combines linearly within a shared representation space, avoiding the operational overhead of hosting two independent autoregressive models.

In parallel, GRACE [21] inserts local, norm-constrained key-value codebook patches directly into intermediate transformer layer activations for lifelong editing.

Whereas GRACE operates within internal hidden representations, RoAS externalizes factual records into measurable atom spaces equipped with explicit Hahn–Jordan cancellation via anti-atoms and certified error-budgeted top- k truncation bounds.

Interpolation between context attention and external memory stores has also been investigated in Memorizing Transformers [22] and adaptive k NN-LM architectures [16]. RoAS complements these formulations by offering a unifying measure-theoretic framework that encompasses token context, dense external corpora, and parameter-expert banks under a common read abstraction with separate norm calibration.

Finally, signed atom spaces interface with the broader machine unlearning literature. Conventional unlearning approaches, such as SISA [23] or Fisher-based certified parameter removal, adjust internal weights to eliminate the statistical influence of target data. In contrast, signed atom spaces ($v = v^+ - v^-$) achieve output-space cancellation strictly within the read operator itself, leaving pretrained parameter weights unmodified.

7.2. Limitations

The atom-space formulation is a shared analytical lens, not a claim that attention, RAG, and MoE systems are algorithmically interchangeable. Synthetic studies use five seeds but remain templated; frozen 1-NN matches the trained read on synthetic insertions. The tri-space and GPT-2 experiments are controlled mechanism studies, and the latter directly mixes retrieved target embeddings into the final hidden state. CounterFact retrieval uses 500 records, while generative steering uses 50 records, and neither is compared end-to-end under matched protocols with SERAC, GRACE, MEMIT [5], MEND [24], WISE [25], or AlphaEdit [26]. The ROME-inspired baseline is not a reproduction of standard ROME. Exact adaptive certification requires computing all pairwise scores; its large selected fractions show it is an error-bound certificate rather than a sublinear algorithm. The finite-dataset Lipschitz diagnostic does not establish a global certificate. Anti-atoms retain both records, cancel only the external read under exact alignment, and do not remove information from dense parameters.

8. Conclusion

We have presented a mathematical formulation casting finite attention, dense retrieval, and expert routing as weighted reads over atom spaces. We derived a conditional retrieval margin result, exact-ranked truncation bounds, an adaptive

415 error-budget rule, and an approximate signed-read cancellation bound, then used
416 controlled studies to diagnose their behavior.

417 Error-budgeted exact truncation provides full empirical coverage while often
418 selecting a large fraction of the corpus. CounterFact reveals strong exact retrieval,
419 where data-calibrated thresholding improves paraphrase activation. The remain-
420 ing studies demonstrate the limitations of templated routing, fragile cancellation
421 under key mismatch, and first-token factual steering in GPT-2 via direct target-
422 embedding injection. Establishing independently certified local retrieval condi-
423 tions, sublinear guarantees, and matched end-to-end evaluations against competi-
424 tive RAG and editing baselines are essential next steps.

425 **CRedit Authorship Contribution Statement**

426 **Farshid Mossaiby:** Conceptualization, Methodology, Formal analysis, Soft-
427 ware, Investigation, Validation, Data curation, Writing - original draft, Writing -
428 review & editing.

429 **Declaration of Competing Interest**

430 The author declares that they have no known competing financial interests or
431 personal relationships that could have appeared to influence the work reported in
432 this paper.

433 **Data Availability**

434 The complete codebase, benchmark definitions, experimental suites, and re-
435 production scripts are publicly available at [https://github.com/mossaiby/](https://github.com/mossaiby/RoAS)
436 [RoAS](https://github.com/mossaiby/RoAS).

437 **Declaration of Generative AI in Scientific Writing**

438 During the preparation of this work, the author utilized Google Gemini to assist
439 in formalizing bounds, experimental pipeline design, and manuscript formatting.
440 After using this tool, the author reviewed and edited the content as needed and
441 takes full responsibility for the content of the publication.

442 References

- 443 [1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez,
444 Ł. Kaiser, I. Polosukhin, Attention is all you need, in: Advances in Neural
445 Information Processing Systems, Vol. 30, 2017, pp. 5998–6008.
- 446 [2] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Nee-
447 lakantan, P. Shyam, G. Sastry, A. Askell, S. Agarwal, A. Herbert-Voss,
448 G. Krueger, T. Henighan, R. Child, A. Ramesh, D. Ziegler, J. Wu, C. Win-
449 ter, C. Hesse, M. Chen, E. Sigler, M. Litwin, S. Gray, B. Chess, J. Clark,
450 C. Berner, S. McCandlish, A. Radford, I. Sutskever, D. Amodei, Language
451 models are few-shot learners, in: Advances in Neural Information Processing
452 Systems, Vol. 33, 2020, pp. 1877–1901.
- 453 [3] M. Geva, R. Schuster, J. Berant, O. Levy, Transformer feed-forward layers are
454 key-value memories, in: Proceedings of the 2021 Conference on Empirical
455 Methods in Natural Language Processing, 2021, pp. 5484–5495.
- 456 [4] K. Meng, D. Bau, A. Andonian, Y. Belinkov, Locating and editing factual as-
457 sociations in GPT, in: Advances in Neural Information Processing Systems,
458 Vol. 35, 2022, pp. 17359–17372.
- 459 [5] K. Meng, A. S. Sharma, A. Andonian, Y. Belinkov, D. Bau, [Mass-Editing](#)
460 [Memory in a Transformer](#) (2023). [arXiv:2210.07229](#).
461 URL <https://arxiv.org/abs/2210.07229>
- 462 [6] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küt-
463 tler, M. Lewis, W. tau Yih, T. Rocktäschel, S. Riedel, D. Kiela, Retrieval-
464 augmented generation for knowledge-intensive NLP tasks, in: Advances in
465 Neural Information Processing Systems, Vol. 33, 2020, pp. 9459–9474.
- 466 [7] U. Khandelwal, O. Levy, D. Jurafsky, L. Zettlemoyer, M. Lewis, [General-](#)
467 [ization through memorization: Nearest neighbor language models](#) (2020).
468 [arXiv:1911.00172](#).
469 URL <https://arxiv.org/abs/1911.00172>
- 470 [8] S. Borgeaud, A. Mensch, J. Hoffmann, T. Cai, E. Rutherford, K. Millican,
471 G. B. V. D. Driessche, J.-B. Lespiau, B. Damoc, A. Clark, D. de Las Casas,
472 A. Guy, J. Menick, R. Ring, T. Hennigan, S. Huang, L. Maggiore, C. Jones,

- 473 A. Cassirer, A. Brock, M. Paganini, G. Irving, O. Vinyals, S. Osindero, K. Si-
 474 monyan, J. W. Rae, E. Elsen, L. Sifre, Improving language models by re-
 475 trieving from trillions of tokens, in: Proceedings of the 39th International
 476 Conference on Machine Learning, Vol. 162, PMLR, 2022, pp. 2206–2240.
- 477 [9] N. Shazeer, A. Mirhoseini, K. Maziarz, A. Davis, Q. Le, G. Hinton, J. Dean,
 478 [Outrageously large neural networks: The sparsely-gated mixture-of-experts](#)
 479 [layer](#) (2017). [arXiv:1701.06538](#).
 480 URL <https://arxiv.org/abs/1701.06538>
- 481 [10] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen,
 482 [LoRA: Low-rank adaptation of large language models](#) (2021). [arXiv:2106.](#)
 483 [09685](#).
 484 URL <https://arxiv.org/abs/2106.09685>
- 485 [11] A. Q. Jiang, A. Sablayrolles, A. Roux, A. Mensch, B. Savary, C. Bam-
 486 ford, D. S. Chaplot, D. de las Casas, E. B. Hanna, F. Bressand, G. Lengyel,
 487 G. Bour, G. Lample, L. R. Lavaud, L. Saulnier, M.-A. Lachaux, P. Stock,
 488 S. Subramanian, S. Yang, S. Antoniak, T. L. Scao, T. Gervet, T. Lavril,
 489 T. Wang, T. Lacroix, W. E. Sayed, [Mixtral of experts](#) (2024). [arXiv:](#)
 490 [2401.04088](#).
 491 URL <https://arxiv.org/abs/2401.04088>
- 492 [12] G. Lample, A. Sablayrolles, M. Ranzato, L. Denoyer, H. Jégou, Large mem-
 493 ory layers with product keys, in: Advances in Neural Information Processing
 494 Systems, Vol. 32, 2019, pp. 8546–8557.
- 495 [13] Y.-H. H. Tsai, S. Bai, M. Yamada, L.-P. Morency, R. Salakhutdinov, Trans-
 496 former dissection: An unified understanding for transformer’s attention via
 497 the lens of kernel, in: Proceedings of the 2019 Conference on Empirical
 498 Methods in Natural Language Processing, 2019, pp. 4344–4353.
- 499 [14] A. F. T. Martins, A. Farinhas, M. Treviso, V. Niculae, P. M. Q. Aguiar,
 500 M. A. T. Figueiredo, Sparse and continuous attention mechanisms, in: Ad-
 501 vances in Neural Information Processing Systems, Vol. 33, 2020, pp. 20989–
 502 21001.
- 503 [15] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, I. Sutskever, Language
 504 models are unsupervised multitask learners, Tech. rep., OpenAI (2019).

- 505 [16] J. He, G. Neubig, T. Berg-Kirkpatrick, Efficient nearest neighbor language
506 models, in: Proceedings of the 2021 Conference on Empirical Methods in
507 Natural Language Processing, 2021, pp. 5703–5714.
- 508 [17] S. Sukhbaatar, A. Szlam, J. Weston, R. Fergus, [End-to-end memory networks](#)
509 (2015). [arXiv:1503.08895](#).
510 URL <https://arxiv.org/abs/1503.08895>
- 511 [18] W. Fedus, B. Zoph, N. Shazeer, Switch transformers: Scaling to trillion pa-
512 rameter models with simple and efficient sparsity, *Journal of Machine Learn-*
513 *ing Research* 23 (120) (2022) 1–39.
- 514 [19] A. Katharopoulos, A. Vyas, N. Pappas, F. Fleuret, Transformers are RNNs:
515 Fast autoregressive transformers with linear attention, in: Proceedings of the
516 37th International Conference on Machine Learning, 2020, pp. 5156–5165.
- 517 [20] E. Mitchell, C. Lin, A. Bosselut, C. D. Manning, C. Finn, Memory-based
518 model editing at scale, in: Proceedings of the 39th International Conference
519 on Machine Learning, 2022, pp. 15817–15831.
- 520 [21] T. Hartvigsen, S. Sankaranarayanan, H. Palangi, Y. Kim, M. Ghassemi, Ag-
521 ing with GRACE: Lifelong model editing with discrete key-value adapta-
522 tions, in: *Advances in Neural Information Processing Systems*, Vol. 36, 2023,
523 pp. 56734–56747.
- 524 [22] Y. Wu, M. N. Rabe, D. Hutchins, C. Szegedy, Memorizing transformers, in:
525 *International Conference on Learning Representations*, 2022.
- 526 [23] L. Bourtole, V. Chandrasekaran, C. A. Choquette-Choo, H. Jia, A. Travers,
527 B. Zhang, D. Lie, N. Papernot, Machine unlearning, in: 2021 IEEE Sympo-
528 sium on Security and Privacy (SP), 2021, pp. 141–159.
- 529 [24] E. Mitchell, C. Lin, A. Bosselut, C. Finn, C. D. Manning, Fast model editing
530 at scale, in: *International Conference on Learning Representations*, 2022.
- 531 [25] P. Wang, Z. Li, N. Zhang, Z. Xu, Y. Yao, Y. Jiang, P. Xie, F. Huang, H. Chen,
532 WISE: Rethinking the knowledge memory for lifelong model editing of large
533 language models, in: *Advances in Neural Information Processing Systems*,
534 Vol. 37, 2024.

- 535 [26] J. Fang, H. Jiang, K. Wang, Y. Ma, S. Jie, X. Wang, X. He, T.-S. Chua, Al-
536 phaEdit: Null-space constrained knowledge editing for language models, in:
537 Proceedings of the 13th International Conference on Learning Representa-
538 tions, 2025.